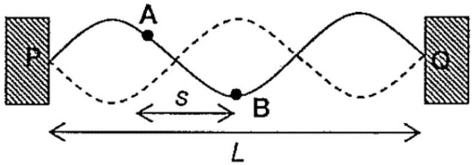
**12** A guitar string of length $L$ is stretched between two fixed points **P** and **Q** and made to vibrate transversely as shown.

**[Turn over**




Two particles **A** and **B** on the string are separated by a distance $s$. The maximum kinetic energies of **A** and **B** are $K_A$ and $K_B$ respectively.

Which of the following gives the correct phase difference and maximum kinetic energies of the particles?

| | Phase difference | Maximum kinetic energy |
|---|---|---|
| **A** | $\left(\frac{3s}{2L}\right)\times 360°$ | $K_A < K_B$ |
| **B** | $\left(\frac{3s}{2L}\right)\times 360°$ | same |
| **C** | 180° | $K_A < K_B$ |
| **D** | 180° | same |

Ans: C
Since particles A and B are at two sides of a node of a stationary wave, they are anti-phase. Hence phase difference is 180°
Maximum KE is proportional to amplitude. Since amplitude of A < amplitude of B,
$K_A < K_B$

**13** Diagram 1 shows a ripple tank experiment in which plane waves are diffracted through a narrow